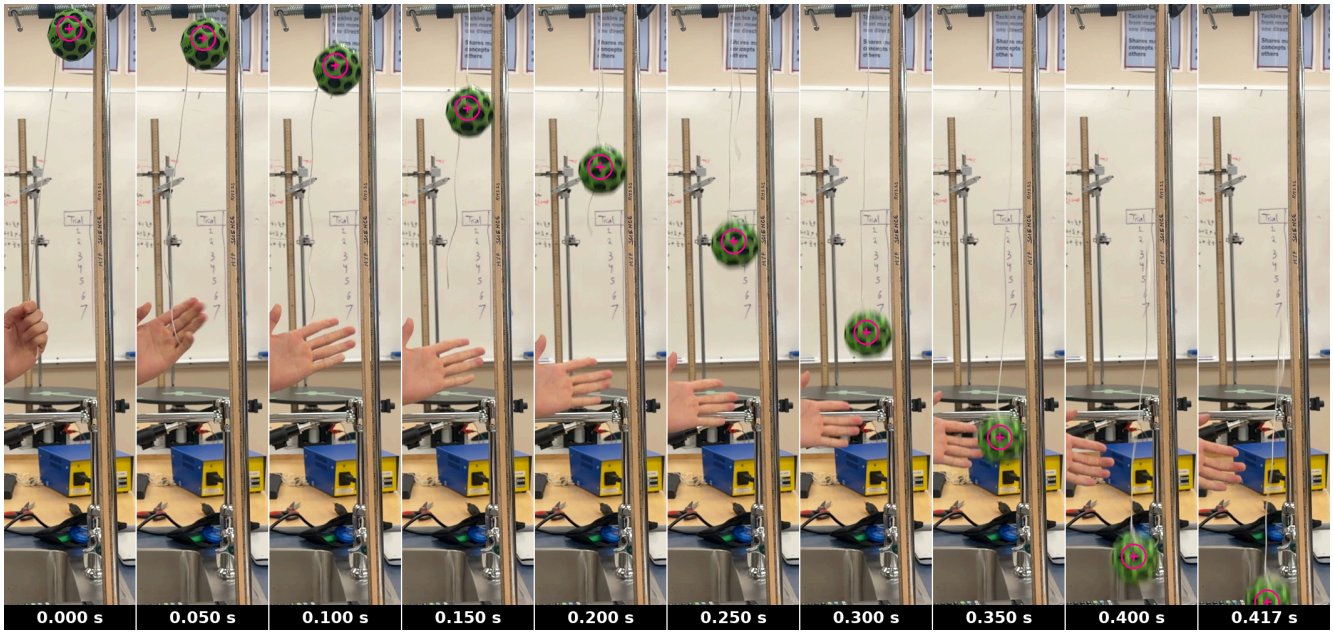
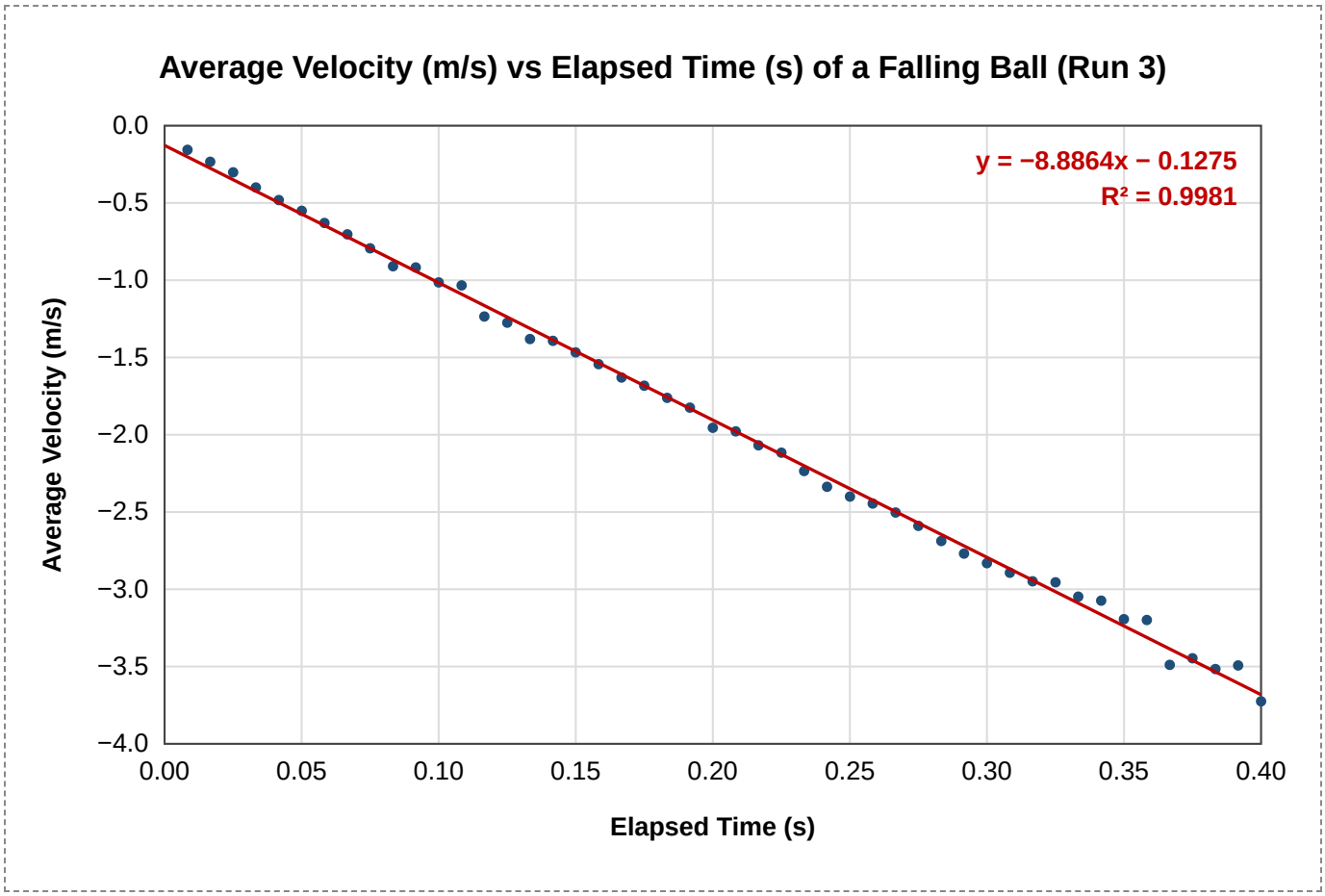
Example of tracking the ball (Run 3, IMG_3051)

Every sixth frame of the fall (120 fps). The pink circle marks the centre of the ball measured in that frame; the time under each frame is the time since the ball was released.



Recorded data for Run 3 (IMG_3051)

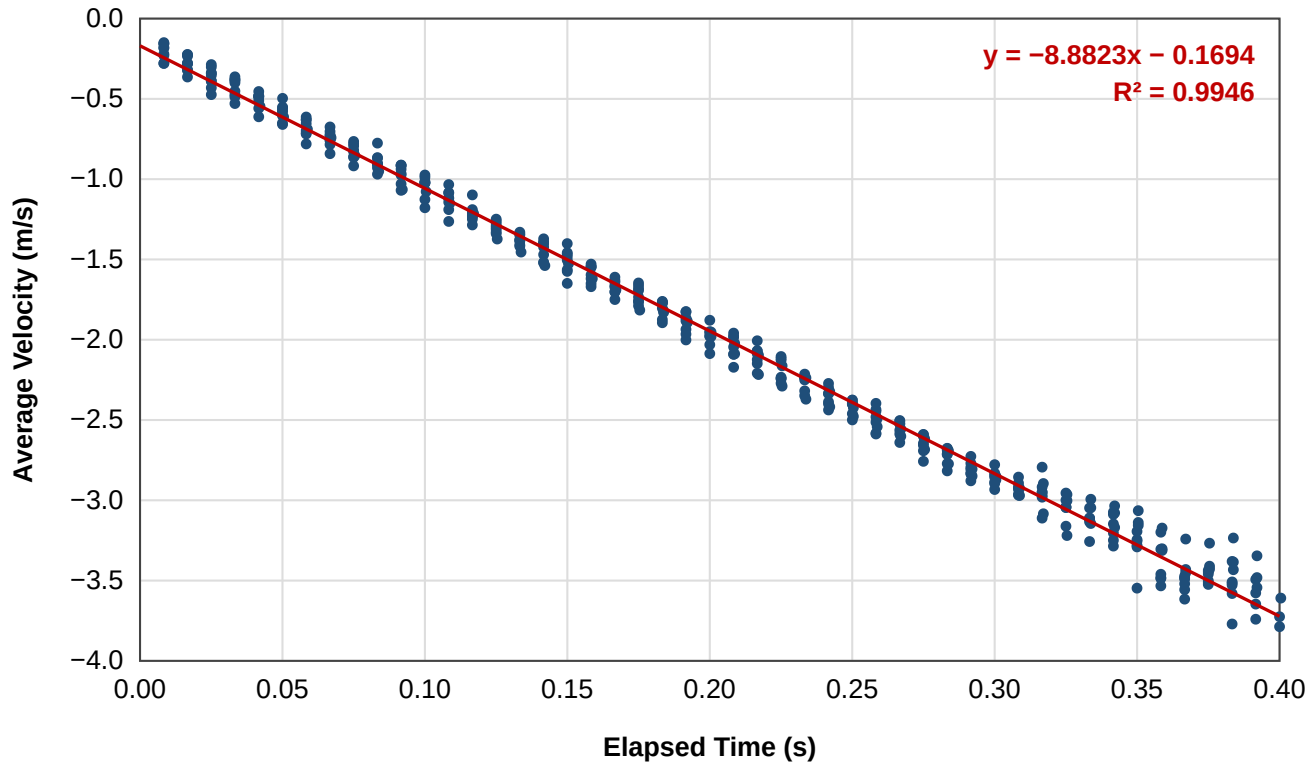
	Time (s)	Height (m)	Smoothed height (m)	Average velocity (m/s)
t_{1}	0.000	0.825		
t_{2}	0.008	0.824	0.824	−0.156
t_{3}	0.017	0.823	0.823	−0.233
t_{4}	0.025	0.821	0.821	−0.302
t_{5}	0.033	0.819	0.818	−0.399
t_{6}	0.042	0.815	0.815	−0.481
t_{7}	0.050	0.811	0.811	−0.551
t_{8}	0.058	0.807	0.806	−0.629
t_{9}	0.067	0.801	0.801	−0.703
t_{10}	0.075	0.795	0.795	−0.794
t_{11}	0.083	0.789	0.789	−0.910
t_{12}	0.092	0.781	0.781	−0.917
t_{13}	0.100	0.773	0.773	−1.014
t_{14}	0.108	0.766	0.765	−1.034
t_{15}	0.117	0.756	0.756	−1.235
t_{16}	0.125	0.747	0.746	−1.274
t_{17}	0.133	0.735	0.735	−1.380
t_{18}	0.142	0.724	0.724	−1.393
t_{19}	0.150	0.712	0.712	−1.467
t_{20}	0.158	0.700	0.700	−1.543
t_{21}	0.167	0.688	0.687	−1.630
t_{22}	0.175	0.674	0.674	−1.683
t_{23}	0.183	0.660	0.660	−1.762
t_{24}	0.192	0.645	0.645	−1.824
t_{25}	0.200	0.630	0.630	−1.955
t_{26}	0.208	0.614	0.613	−1.977
t_{27}	0.217	0.597	0.597	−2.068
t_{28}	0.225	0.580	0.580	−2.116
t_{29}	0.233	0.562	0.562	−2.235
t_{30}	0.242	0.544	0.543	−2.336
t_{31}	0.250	0.524	0.524	−2.400
t_{32}	0.258	0.504	0.504	−2.446
t_{33}	0.267	0.484	0.484	−2.503
t_{34}	0.275	0.463	0.463	−2.589
t_{35}	0.283	0.441	0.441	−2.688
t_{36}	0.292	0.419	0.419	−2.769
t_{37}	0.300	0.396	0.396	−2.831
t_{38}	0.308	0.372	0.372	−2.893
t_{39}	0.317	0.348	0.348	−2.948
t_{40}	0.325	0.324	0.323	−2.955
t_{41}	0.333	0.298	0.299	−3.048
t_{42}	0.342	0.274	0.273	−3.073
t_{43}	0.350	0.247	0.248	−3.194
t_{44}	0.358	0.222	0.221	−3.199
t_{45}	0.367	0.194	0.194	−3.489
t_{46}	0.375	0.167	0.165	−3.446
t_{47}	0.383	0.134	0.137	−3.517
t_{48}	0.392	0.108	0.107	−3.492
t_{49}	0.400	0.080	0.078	−3.726
t_{50}	0.408	0.047	0.047	
t_{51}	0.417	0.015		


Average Velocity (m/s) vs Elapsed Time (s) of a Falling Ball (Run 3)
y = −8.8864x − 0.1275
R² = 0.9981
Average Velocity (m/s)
Elapsed Time (s)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40

Value of g for each run

Run #	Video	Trendline equation	R^2 value	Value of g (m/s²)
1	IMG_3049	y = −8.6225x − 0.2339	0.9901	8.6225
2	IMG_3050	y = −8.7772x − 0.1657	0.9980	8.7772
3	IMG_3051	y = −8.8864x − 0.1275	0.9981	8.8864
4	IMG_3052	y = −8.6806x − 0.1454	0.9971	8.6806
5	IMG_3053	y = −9.0588x − 0.1516	0.9985	9.0588
6	IMG_3054	y = −9.1246x − 0.0827	0.9982	9.1246
7	IMG_3055	y = −8.8489x − 0.2167	0.9982	8.8489
8	IMG_3056	y = −8.9518x − 0.1558	0.9975	8.9518
9	IMG_3057	y = −9.0938x − 0.2318	0.9983	9.0938

g = −(slope of the average velocity vs elapsed time trendline). Mean = 8.8938 m/s², sample standard deviation (TI-84 Sx) = 0.1798 m/s², population (TI-84 σx) = 0.1695 m/s², n = 9.



One straight line through every velocity point from all 9 runs: g = 8.882 ± 0.032 m/s² (standard error of the slope).

Air resistance check

g from the first half and the second half of each fall (slope of each half of the velocity–time data).

Run #	g, first half (m/s²)	g, second half (m/s²)
1	9.34	7.11
2	9.13	8.19
3	9.22	8.53
4	9.11	8.07
5	9.15	8.94
6	9.43	9.10
7	9.21	8.33
8	9.22	8.51
9	9.17	8.99
Mean	**9.22**	**8.42**